



Supply chain risk: Changes in supplier composition and vertical integration

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ABSTRACT

Using textual analysis of earnings conference calls, we quantify firms' supply chain risk and explore how firms react when supply chain risk increases. We show that firms with supply chains that span across continents, multinationals, and firms with fewer suppliers of an input face higher supply chain risk. In addition, firms exhibit high supply chain risk when their suppliers also do so. Firms manage supply chain risk by establishing relationships with closer and domestic suppliers and with suppliers that are industry leaders and by vertically integrating, but they continue to work with suppliers in other continents.

1. Introduction

Production relies on global and complex supply chains, which have often been optimized to reduce costs. Major events, such as the Sino-American trade war, the Covid-19 pandemic, the Suez Canal accident, and the 2011 Japanese earthquake, disrupt supply chains and production. Existing literature has widely documented that even small negative shocks, such as bankruptcies or natural disasters, are transmitted to firms upstream and downstream (Hertzel et al., 2008; Acemoglu et al., 2012; Barrot and Sauvagnat, 2016; Carvalho et al., 2021). However, we know little about which firms are more exposed or even whether firms systematically update their investors about the ex-ante probability that such shocks may occur and any perceived changes in this source of risk. We are also unable to quantify the effects of supply chain risk on corporate policies. Concerns have been raised that firms are unaware of the supply chain risks their suppliers are subject to (Choi et al., 2020) and may not be able to manage this source of risk.

Quantification of the supply chain risk faced by different firms would be important but is challenging because supply chain risk comes from many sources and multiple channels. For instance, firms may be indirectly exposed if their suppliers, or the suppliers of

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their suppliers, face bottlenecks. Similarly, firms may be exposed through their customers if downstream firms are unable to source complementary inputs and are forced to limit production. Furthermore, commercial data sources mainly focus on big customers and suppliers, providing limited coverage on the potential source of shocks over the supply network. While these data sources are useful for quantifying the effects of shock propagation, they do not allow us to gauge how firms perceive supply chain risk and adapt their strategies to mitigate supply assurance concerns.

Our objective in this paper is to develop a proxy for supply chain risk using textual analysis, and to study which firms are most affected by supply chain risk and the extent to which supply chain risk affects firms' policies and industrial structure. We perform textual analysis of earnings conference calls to construct a measure of supply chain risk faced by U.S. listed companies. Following Hassan et al. (2019), we measure supply chain risk using the discussion of supply chain issues related to words capturing risk and uncertainty. We also use topic modeling analysis to ascertain the sources of supply chain risk and construct a measure of the sentiment of supply chain discussions to capture negative realizations of supply chain shocks. Supply chain sentiment can also help us capture news about the conditional mean of supply chain shocks (see Hassan et al. (2019, 2023a, 2023b)) for a similar interpretation of political sentiment).

We find that between 2002 and 2020 most U.S. companies discuss topics related to supply chains in connection to risk and uncertainty. This indicates that supply chain risk is important and so far neglected in the economics and finance literature. Importantly, supply chain risk is positively correlated with stock price volatility, while supply chain sentiment is associated with positive returns. Consistent with studies that highlight that measured uncertainty in aggregate data tends to increase when the economy is affected by adverse shocks (Berger et al., 2020), supply chain sentiment turns negative and supply chain risk increases on average in conjunction with events that are known to have disrupted supply chains, such as the 2011 Japanese earthquake and the Thai floods (Carvalho et al., 2021). The increase in supply chain risk and the drop in supply chain sentiment appear unprecedented during the Covid-19 pandemic. However, while supply chain risk and supply chain sentiment are negatively correlated, the actual correlation across firms over time is only -1% , suggesting that we can independently measure negative realizations of supply chain shocks and supply chain uncertainty. Furthermore, supply chain risk appears to be higher for firms in industries that use differentiated products as inputs, consistent with the intuition that these goods are hard to substitute.

Even though macroeconomic and industry level uncertainty matters, the way firms discuss supply chain risk appears to be highly idiosyncratic. Most of the variation in supply chain risk is explained by firm-specific shocks rather than time- or industry-specific shocks. We provide a battery of tests showing that this idiosyncratic variation is unlikely to be noise and that it largely depends on firm characteristics affecting exposure to supply chain risk. Supply chain risk is higher for firms that have suppliers in different continents and are small relative to their suppliers, suggesting that they have limited bargaining power. Firms that have many suppliers in a given industry are less exposed to supply chain risk, suggesting that hold-up problems and lack of diversification in input sources magnify supply chain uncertainty. Large firms and multinationals, possibly having more complex supply chains, are more exposed to supply chain risk.

We also investigate the actions firms take to manage supply chain risk. First, firms appear to actively manage supply chain risk by increasing the number of their suppliers. In addition, firms that communicate more uncertainty about their supply chain subsequently establish relationships with suppliers that can be considered industry leaders and with nearby suppliers, suggesting that these firms attempt to increase the reliability of their supplier network. However, we find no evidence that firms sever their relationships with suppliers in other continents, suggesting that supply chain diversification and nearshoring do not necessarily imply more fragmentation.

In addition, we find that supply chain risk affects the boundaries of the firm and industrial structure. The benefits of common ownership of different stages of the production process are expected to increase when there is uncertainty about the availability of inputs (Williamson, 1971). Accordingly, firms that report high supply chain risk are involved in more vertical mergers and acquisitions (M&As). This indicates that firms tend to acquire customers and suppliers when supply chain risk increases. Financial constraints limit firms' ability to perform M&As, potentially hampering their long-term competitive advantage.

Interestingly, changes in corporate policies are driven by supply chain risk and not by supply chain sentiment. Supply chain sentiment, which is presumably more closely related to the negative shocks affecting a firm's suppliers, has different or no effects on supplier composition and vertical M&As. This is consistent with Ersahin et al. (2023), who find that supply chains are overall stable following negative shocks and suggest that financially flexible firms react when uncertainty increases, not when negative shocks occur.

Our results are robust to a battery of tests and alternative specifications. First, we address the concern that firms may discuss supply chain risk to justify vertical M&As and changes in their supplier composition, not because they are experiencing actual uncertainty in sourcing inputs. To do so, we instrument for our textual measure of supply chain risk using the supply chain risk experienced by a firm's suppliers, which can be considered exogenous to a firm's plans. Consistent with our main findings, we document increases in vertical M&As as well as in the number of suppliers that are geographically closer or industry leaders, when we use exogenous variation in a firm's supply chain risk arising from its suppliers.

Second, the results are robust if we control for firm-level measures of political risk and climate risk, which are constructed with similar techniques and have been shown in previous literature to affect firms' policies (Hassan et al., 2019, 2023a, 2023b; Sautner et al., 2023). These findings indicate that our measure of supply chain risk captures a different source of shocks and uncertainty. More importantly, while firms appear to decrease investment when they face political risk and other sources of macroeconomic uncertainty (Baker et al., 2023), they increase investment by acquiring customers and suppliers when uncertainty arises from the availability of inputs, suggesting that supply chain risk deserves independent investigation.

Last, we show that our results are invariant whether we control for supply chain sentiment or not and when we control for the overall sentiment of the conference call, alleviating any concerns that supply chain risk may be correlated with negative shocks or with changes in the conditional mean of supply chain shocks.

Our work is related to several strands of the literature. First, we contribute to the literature on the boundaries of the firm. Production is believed to be coordinated within a firm, rather than through the market when transaction costs and hold-up problems are severe (Coase, 1937; Klein et al., 1978; Grossman and Hart, 1986). A more recent strand of this literature focuses on global supply chains and explores the effects of demand and technological characteristics on the decision to integrate different stages of production (e.g., Antràs and Chor, 2013; Alfaro et al., 2019), but neglects the effects of risk. While it is well recognized that mergers facilitate collaboration especially in innovative industries (Bena and Li, 2014; Hsu et al., 2022), there are few empirical studies exploring vertical mergers. Existing work focuses on the role of industry shocks (Ahern and Harford, 2014), cash-flow uncertainty (Garfinkel and Hankins, 2011), and R&D incentives (Fan and Goyal, 2006; Frésard et al., 2020).

The role of supply chain risk in vertical integration decisions, first highlighted by Williamson (1971), has been largely neglected in subsequent investigations of vertical M&As, even though theoretically, supply assurance concerns are known to affect the decision to integrate vertically (Bolton and Whinston, 1993). To the best of our knowledge, we are the first to show empirically that supply chain risk is a driver of vertical integration.

Second, from a methodological point of view, we contribute to a nascent literature that uses textual analysis to measure risk and uncertainty. The most prominent contributions relate to measures of political risk (Baker et al., 2016; Hassan et al., 2019). Textual analysis has also been widely used in measuring geopolitical risk (Caldara and Iacoviello, 2022), climate risk and climate risk disclosure (Sautner et al., 2023; Li et al., 2023), and cyber risk (Florackis et al., 2023). We construct a measure of supply chain risk based on textual analysis and show that this source of risk warrants independent investigation not only because it is not subsumed by earlier proxies of uncertainty, but also because it has different effects on corporate policies.

Finally, the literature on networks highlights the importance of complementarities between different phases of the production process (Kremer, 1993). While empirical work typically studies the consequences of negative realizations of supply chain risk, recent theories acknowledge that companies' decisions to diversify the source of inputs reflect disruption risk in a way that may lead to a decline in output and is not necessarily socially optimal (Bimpikis et al., 2018; Kopytov et al., 2022). To the best of our knowledge, we are the first to explore how firms manage their supply chains in response to this source of risk. This contrasts with previous literature that typically takes the supply chain as given and explores how shocks are transmitted given current customer-supplier links focusing on natural disasters (e.g., Barrot and Sauvagnat, 2016; Carvalho et al., 2021), credit shocks (Alfaro et al., 2021; Costello, 2020), data breaches (Crosignani et al., 2023), or pandemic closures (Bonadio et al., 2021; Aral et al., 2021). Using our newly developed proxy for supply chain risk, we capture firm-level perceived supply chain uncertainty potentially arising from any of the above shocks, while controlling for supply chain negative shock realizations using supply chain sentiment and other proxies.

2. Data sources

We combine a variety of data sources. First, we collect about 200,000 transcripts of conference calls held in conjunction with an earnings release (hereafter, earnings calls) by 8000 public firms listed in the United States from 2002 to 2020 through Refinitiv Eikon database and Capital IQ. Firms generally hold quarterly earnings conference calls to inform investors and analysts about the firm's performance. Presentation by the management is followed by a question-and-answer session. Conference calls have been widely used

Table 1
Summary Statistics.

	(1)	(2)	(3)	(4)	(5)	(6)
Variables	N	Mean	SD	P25	P50	P75
SCRisk	36,134	4.4782	8.3637	1.2223	2.1667	4.0098
SCSentiment	36,134	40.9988	95.5976	3.8604	16.8212	42.4933
Different continents	36,134	0.1383	0.2415	0	0	0.2000
Relative size	36,134	0.7752	8.1713	0.0069	0.0391	0.3424
Average number of suppliers in an input industry	36,134	1.4874	0.8856	1.0000	1.1538	1.5600
Multinational	36,134	0.6502	0.4769	0	1	1
Customer dependence	36,134	0.1907	0.3928	0	0	0
Market share	36,134	0.0143	0.0379	0.0002	0.0013	0.0084
Financial constraint	36,134	0.3417	0.4743	0	0	1
Institutional ownership	36,134	0.5625	0.3651	0.2185	0.6752	0.8708
Size	36,134	6.7552	1.8589	5.4282	6.6804	7.9717
Tobin's Q	36,134	2.1986	1.5636	1.2287	1.6796	2.5610
Cash holdings	36,134	0.2256	0.2359	0.0448	0.1349	0.3324
Cash flow	36,134	-0.0182	0.2063	-0.0251	0.0365	0.0775
Number of suppliers	36,134	7.9352	11.5148	1	3	9
Number of suppliers in the same continent	36,134	3.7795	6.1158	0	1	4
Number of U.S. suppliers	36,134	3.6500	5.8404	0	1	4
Number of suppliers in different continents	36,134	1.3657	2.6310	0	0	1
Number of industry leader suppliers	36,134	3.6166	5.3439	0	1	5
M&A with supplier	36,134	0.0991	0.2988	0	0	0
M&A with customer	36,134	0.0720	0.2585	0	0	0
Unrelated M&As	36,134	0.0380	0.1911	0	0	0

Notes: This table presents summary statistics for the main variables used in our analysis. All variables are defined in Internet Appendix A.

to construct proxies for corporate culture (Li et al., 2021b) as well as to quantify firms' exposure to political risk (Hassan et al., 2019), Covid-19 (Hassan et al., 2023a), Brexit (Hassan et al., 2023b), cyber risk (Florackis et al., 2023), and climate risk (Sautner et al., 2023; Li et al., 2023). We construct our proxies for supply chain risk and sentiment using the entire conference call including both the presentation and the question-and-answer session, following the approach introduced by Hassan et al. (2019).

Second, we obtain information on supply chains from Factset Revere, including specific supplier-customer pairs and their locations. Factset Revere collects relationship information from primary public sources, such as SEC 10-K annual filings, investor presentations, and press releases, and classifies the relationship types. Factset Revere spans the period 2002–2020 and limits our sample period. On average, for the sample firms, we observe 8 suppliers and 10 customers.

Third, we use the Securities Data Company (SDC) Mergers and Acquisitions database for M&As. To identify whether the target and the acquirer are in vertically related industries, we use the Bureau of Economic Analysis's (BEA) Input-Output tables, which provide the dollar flow of goods and services between producing and purchasing industries.

Finally, we use Compustat and CRSP for firm-level variables. Table 1 summarizes the main variables that we introduce as we progress with the analysis. All variables are defined in Internet Appendix A.

3. Measuring supply chain sentiment and risk

We view supply chain risk to arise from any shocks to a firm's production network. Hence, supply chain risk may arise because uncertainty about future input and commodity prices, tariffs, changes in transportation costs, uncertainty about offshore activities and bottlenecks affect the timely delivery of high-quality inputs or the probability of securing inputs at a price consistent with current profit margins. In addition, geopolitical risk may disrupt direct and indirect customer or supplier relationships, affecting firms' sales or ability to produce.

Overall, supply chain risk may arise from a variety of sources that are hard to quantify using financial data: Not only firms have thousands of customers and suppliers that are typically hard to observe, but even if we observed all the shocks affecting a firm's production network, we would be unable to quantify a firm's exposure because suppliers may choose to prioritize different customers and markets. For these reasons, we conduct textual analysis to quantify the supply chain risk to which a firm is exposed using managerial statements in conference calls.

3.1. Textual analysis

We build on recent work that uses the proportion of the conversations during a conference call that is centered on a particular topic as a source for identifying various risks and opportunities (Hassan et al., 2019, 2023a, 2023b). To construct our dictionary, we adopt methods developed in computational linguistics and natural language processing to capture word combinations that are indicative of discourse about a given topic (Song and Wu, 2008; Manning et al., 2008). Specifically, we follow Hassan et al. (2019), who study firm-level political risk and determine signal bigrams by comparing training libraries of a political textbook to bigrams appearing in nonpolitical texts, specifically an accounting textbook.

We thus construct a training library of bigrams related to supply chains using the textbook, *Supply Chain Management: Strategy, Planning, and Operation* (6th edition; Chopra and Meindl, 2016). The textbook, born from a course on supply chain management taught at Northwestern University's Kellogg School of Management, focuses on the economics and management of supply chains, rather than on the mathematics of inventory optimization, and ranks among the top books in supply chain management in a variety of reviews.¹ We also construct another training library of non-supply-chain topics using the financial accounting textbook, *Financial Accounting* (10th edition; Libby et al., 2020), which allows us to capture words used in the discussion of general finance and accounting issues. We define the training library archetypical of the discussion of supply chain issues, S , and the other training library of general corporate financial issues, N . Each training library is the set of all adjacent two-word combinations (bigrams) contained in the respective supply chain and financial accounting texts.

We consider as related to supply chain issues all bigrams that appear in the supply chain textbook but not in the financial accounting textbook; in addition, since there is some overlap between supply chain and financial accounting topics, and the bigram "supply chain" naturally appears in the financial accounting textbook, we include in our supply chain training library any bigrams that are at least 30 times more frequent in the supply chain textbook than in the financial accounting textbook.

Using this methodology, we identify a total of 70,820 bigrams associated with supply chain discussions, of which only 59 also appear in the financial accounting textbook, but are at least 30 times more frequent in the supply chain book. The dictionary we obtain in this way includes words that are not easily relatable to supply chain discussions, such as "in table", "in period", etc. For this reason, we manually clean the resulting dictionary and remain with 4631 bigrams. Table IA.1 lists the top 100 supply chain bigrams in the training library. "Demand Uncertainty," "Production Cost," and "Transportation Cost(s)" figure among the top 100 bigrams.

Interestingly, almost all firms discuss supply chain topics, indicating that this is an important issue, so far largely neglected in the literature. Having identified the relevant bigrams, we decompose all the conference calls of firm i in year t into a list of bigrams contained in the filings, $b = 1, \dots, B_{it}$, and assign a score to each quarterly earnings call transcript based on the recurrence of bigrams in our dictionary.

¹ See for instance <https://insights.btoes.com/top-10-supply-chain-books>, and <https://www.linkedin.com/pulse/top-4-supply-chain-books-every-student-should-read-mohammed-boulalam/>

We construct our scores at a yearly frequency because firms are likely to face switching and search costs when changing suppliers or deciding whether to vertically integrate. Since any reactions are likely to take more than a quarter, measuring supply chain risk at a higher frequency may just increase noise. To define supply chain risk, we count the number of occurrences of bigrams indicating discussion of supply chains within the set of 10 words surrounding a synonym for “risk” or “uncertainty” on either side in the earnings calls performed during year t , and divide by the total number of bigrams in the transcript:

$$SCRisk_{i,t} = \frac{\sum_b^{B_{i,t}} I[b \in S \setminus N] \times I(|b - r| < 10) \times \frac{f_{b,S}}{B_S}}{B_{i,t}},$$

where $I[\bullet]$ is the indicator function, r is the position of the nearest synonym of risk or uncertainty, $f_{b,S}$ is the frequency of the term b in the supply chain training library, and B_S is the total number of terms in the supply chain training library. The numerator thus simply counts the number of bigrams associated with discussion of supply chains that occur within ten words to a synonym for risk or uncertainty. Terms are given a larger weight if they recur in the training library more often. The denominator, $B_{i,t}$, is the total number of bigrams in the transcript of firm i in year t . Table IA.2 reports the list of synonyms of risk words derived from the Oxford Dictionary following Hassan et al. (2019).

A challenge for any measure of risk and uncertainty is that news about the variance of shocks may be correlated with negative shocks or shocks to their conditional mean (e.g., Bloom, 2014). For this reason, following a procedure similar to that for the construction of *SCRisk*, we construct a proxy for the sentiment of the supply chain discussions. Specifically, we condition on the proximity to positive and negative words, identified from Loughran and McDonald’s (2011) dictionary of words related to sentiment in financial texts. Supply chain sentiment is thus defined as:

$$SCSentiment_{i,t} = \frac{\sum_b^{B_{i,t}} \left\{ I[b \in S \setminus N] \times \sum_{c=b-10}^{b+10} S(c) \times \frac{f_{b,S}}{B_S} \right\}}{B_{i,t}},$$

where $S(c)$ is a function that assigns a value of +1 if bigram c is associated with positive sentiment and a value of −1 if bigram c is associated with negative sentiment; $S(c)$ takes value zero otherwise. $\sum_{c=b-10}^{b+10} S(c)$ calculates the net sentiment among the ten words surrounding bigram b . Also, in this case, we weigh words based on their frequency in the training library.

Based on existing literature (Hassan et al., 2019, 2023a, 2023b), we view *SCSentiment* and *SCRisk* as proxies for the first and the second moment of a firm’s supply chain shocks during a year, respectively. However, the interpretation of our findings is unchanged if we, more informally, consider *SCRisk* as uncertainty and fear of future supply chain shocks, and *SCSentiment* as capturing the realization of shocks related to the supply chain during a year; in this case, to the extent that supply chain shocks are correlated over time for the same firm, *SCSentiment* could also capture the conditional mean of supply chain shocks.²

Naturally, given that measures of uncertainty typically increase when adverse shocks occur (Berger et al., 2020; Bloom, 2014), *SCRisk* and *SCSentiment* are negatively correlated. When a firm receives news that the provision of some inputs is disrupted, it simultaneously faces higher uncertainty on the future stability of its supply chain. However, the correlation between *SCSentiment* and *SCRisk* is low at around −1%, indicating that these two proxies have independent sources of variation.

3.2. What is supply chain risk about?

To provide an overview of the determinants of supply chain risk, we use topic modeling analysis, a widely used machine learning algorithm (see, e.g., Li et al., 2021a; Dasgupta et al., 2020), on snippets containing words in the risk and supply chain dictionaries. Applying this algorithm, we obtain several lists of words that tend to occur jointly and can therefore be considered related to different topics. From these lists, which Fig. 1 presents as word clouds, we can infer the topics associated with discussions of supply chain risk. We assign meaningful labels to the topics by inspecting the list of words.

Supply chain risk appears to be linked to uncertainty about commodity price and input costs, technology and cyberattacks, the environment, climate risk, and the pandemics. Firms also discuss supply chain risk in conjunction with investment and acquisitions, suggesting that firms manage this source of risk by updating their lines of production to accommodate new suppliers and even consider vertical integration. Finally, firms discuss supply chain risk in conjunction with liquidity, difficult access to loans, and other financial

² Our interpretation is confirmed by the excerpts of earning conference calls that score highest for *SCRisk* and *SCSentiment* in Panels A and B, respectively, of Table IA.3. To further evaluate our interpretation of the textual analysis proxies, we also use the dictionary of Bozanic et al. (2018) to capture future statements. Even though this dictionary is adapted to disclosures rather than earnings conference calls (and therefore we may not identify all future statements), we find that 75% of the snippets that we identify as discussing supply chain risk include future statements, which is overall supportive of our interpretation of *SCRisk*, as uncertainty can only be related to future events. Only 50% of the snippets discussing supply chains with positive or negative sentiment contain forward-looking statements, which is consistent with the fact that *SCSentiment* can be backward-looking (if shocks that already occurred are being discussed) or forward-looking (if firms express negative expectations about future realizations of supply chain shocks).

which are presumably hard to substitute, tend to have higher supply chain risk (but not lower supply chain sentiment). By converse, firms that are less dependent on their suppliers because they are in upstream industries (produce crude products) have lower SCRisk on average. This finding suggests that uncertainty originating from suppliers is more relevant for a firm than that from customers.

Fig. IA.1 describes how the means of SCRisk and SCSentiment vary over time. Supply chain risk appears to be heightened and, to a somewhat lower extent, supply chain sentiment becomes more negative in connection to events that are widely known to have disrupted global supply chains, such as the 2011 great East Japan earthquake and the Thailand floods, the Sino-American trade war, and more recently the Covid-19 outbreak.⁴ Supply chain sentiment oscillates between negative and positive.

While the variation of SCRisk across time and industries is meaningful, only a limited proportion of supply chain risk is explained by time or industry specific shocks. In Table IA.7, economy-wide shocks, as captured by time fixed effects, industry fixed effects (considered at different levels of granularity) and, more relevantly, interactions of industry and time fixed effects and even firm fixed effects explain at most about 50% of SCRisk (and SCSentiment), suggesting that firm-specific shocks are important. Importantly, Panel A of Table IA.8 shows that firms with higher SCRisk tend to have higher realized volatility. Furthermore, supply chain sentiment is positively associated with returns (Panel B), indicating that we capture actual idiosyncratic firm-level performance volatility.

4. Firms' exposure to supply chain risk

Table 2 relates SCRisk to contemporaneous firm characteristics to understand which firms are relatively more exposed. Throughout the analysis, we control for interactions of industry and year fixed effects. The estimates indicate that supply chain risk is higher for firms that report a higher fraction of suppliers in other continents and large firms that tend to have more complex supply chains. A one-standard-deviation increase in the fraction of suppliers located in other continents is associated with a 5.0% increase in our measure of supply chain risk relative to the sample median. Similarly, a one-standard-deviation increase in firm size is associated with a 6.6% increase in supply chain risk. Multinationals also tend to have higher supply chain risk. These findings suggest that distance and supply chain complexity increase supply chain risk.

Supply chain risk also appears to be related to a company's bargaining power with its suppliers. First, companies that are large relative to their suppliers exhibit less supply chain risk, suggesting that they may benefit from being the most valued customers of their suppliers. Second, firms that have multiple providers of the same input should be better able to substitute suppliers. We thus consider firms with a higher average number of suppliers by input industry. These firms have lower SCRisk: On average, one more supplier by input industry decreases a firm's supply chain risk by 8.9% relative to the sample median, suggesting that diversification decreases supply chain risk.

While in principle supply chain risk may also arise from uncertainty about customers, we do not observe that customer dependence, which should presumably increase a firm's vulnerability, is positively associated with supply chain risk. Financial constraints also do not appear to affect supply chain risk, while firms with higher growth opportunities, which possibly rely more on knowledge and services than physical inputs for their production, face less supply chain risk. Finally, it does not appear that institutional owners push firms to discuss supply chain issues and risk, as institutional ownership is negatively related to our proxy for supply chain risk.

Interestingly, supply chain sentiment (reported in Table IA.9), but not supply chain risk, is more negative for small firms and firms that face a more competitive environment as proxied by a low market share, suggesting that these firms have more unstable supply chains. The estimated coefficient is not only statistically significant but also economically significant: a 1% decrease in market share is associated with a 3.4% decrease in supply chain sentiment.

5. How firms manage supply chain risk

Since the early 1980s, production has been relying on complex global value chains (Antràs and Chor, 2022). Geopolitical tensions between the U.S. and China, the war in Ukraine, and the Covid-19 pandemic are increasingly mentioned in the press as leading to more fragmentation and reshoring of global supply chains. Systematic evidence is however missing. In principle, firms could achieve more resilience by diversifying their suppliers, rather than relying on closer suppliers. Using our proxy for firm level supply chain risk, we can provide the first systematic evidence about how firms organize their supply chains and their production processes in response to an increase in supply chain risk.

5.1. Supplier composition

We investigate how characteristics of the supply chain vary when firms face heightened supply chain risk. Specifically, we relate characteristics of a firm's supply chain to the one-year lag of the proxy for supply chain risk, while including firm fixed effects and interactions of industries and year fixed effects. In this way, we study how changes in supply chain risk affect how firms decide to source their inputs. Throughout the table, we control for supply chain sentiment and control for a firm's cash flow, among other controls, at $t-1$ to capture the effect of negative shocks that propagate over the supply chain.

Table 3 shows that firms attempt to reduce the probability of a supply chain disruption by changing their supplier pool. Bimpikis et al. (2018) and Elliott et al. (2022) predict that firms may react to heightened supply chain risk by increasing the number of suppliers

⁴ Figure IA.2 relates the yearly mean of SCRisk with a measure of supply chain strains based on transportation costs, developed by Benigno et al. (2022), and confirms that the two measures evolve similarly.

Table 2
Firm Characteristics and Supply Chain Risk.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	SCRisk	SCRisk	SCRisk	SCRisk	SCRisk	SCRisk	SCRisk	SCRisk	SCRisk	SCRisk
Different continents	0.6834** (0.2687)	0.6834** (0.2687)	0.5184** (0.2637)	0.5281** (0.2635)	0.4780* (0.2635)	0.4782* (0.2634)	0.4760* (0.2636)	0.4647* (0.2639)	0.4371* (0.2649)	0.4432* (0.2641)
Relative size		−0.0032* (0.0018)	−0.0061*** (0.0019)	−0.0052*** (0.0017)	−0.0050*** (0.0016)	−0.0051*** (0.0016)	−0.0053*** (0.0017)	−0.0053*** (0.0017)	−0.0054*** (0.0017)	−0.0055*** (0.0017)
Size			0.1326*** (0.0373)	0.1791*** (0.0415)	0.1443*** (0.0424)	0.1440*** (0.0424)	0.1340*** (0.0450)	0.1723*** (0.0483)	0.1970*** (0.0504)	0.1671*** (0.0509)
Average number of suppliers industry				−0.2169*** (0.0652)	−0.2002*** (0.0654)	−0.1998*** (0.0652)	−0.2139*** (0.0658)	−0.2168*** (0.0659)	−0.2287*** (0.0661)	−0.2058*** (0.0662)
Multinational					0.4393*** (0.1279)	0.4395*** (0.1279)	0.4332*** (0.1281)	0.4167*** (0.1280)	0.4180*** (0.1281)	0.3982*** (0.1276)
Customer dependence						−0.0324 (0.1357)	−0.0309 (0.1356)	−0.0478 (0.1366)	−0.0285 (0.1370)	−0.0372 (0.1364)
Market share							0.0176 (0.0186)	0.0146 (0.0186)	0.0116 (0.0186)	0.0171 (0.0186)
Financial constraint								0.3484*** (0.1323)	0.3174** (0.1321)	0.2941** (0.1320)
Institutional ownership									−0.4143** (0.1664)	−0.3164* (0.1683)
Tobin's Q										−0.1741*** (0.0366)
Ind x year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	36,043	36,043	36,043	36,043	36,043	36,043	36,043	36,043	36,043	36,043
Adj. R-squared	0.1070	0.1070	0.1077	0.1080	0.1084	0.1084	0.1084	0.1085	0.1087	0.1096

Notes: This table relates SCRisk to contemporaneous firm characteristics in an annual panel. The main independent variable in column (1) is *Different continents*, which is the fraction of a firm's suppliers who are located in a different continent over the total number of suppliers. The additional independent variable in column (2) is *Relative Size*, defined as a firm's total assets divided by the average total assets of its suppliers. The additional independent variable in column (3) is *Size*, defined as the natural logarithm of the firm's total assets. The additional independent variable in column (4) is *Average number of suppliers by industry*, defined as the average of a firm's number of suppliers within each of the three-digit SIC industries for which we observe suppliers from Factset Revere. The additional independent variable in column (5) is *Multinational*, defined as an indicator variable that equals one if the firm has geographical segments in other countries based on Compustat segment data. The additional independent variable in column (6) is *Customer dependence*, defined as an indicator variable that equals one if the firm reports major customers that purchase at least 10% of its sales in Compustat segment data. The additional independent variable in column (7) is *Market Share*, defined as a firm's sales divided by the total sales in the firm's industry, which is scaled up by a factor of 100. The additional independent variable in column (8) is *Financial constraint*, which is an indicator variable that equals one if the [Whited and Wu \(2006\)](#) proxy for firm-level financial constraints is above the median of our sample. The additional independent variable in column (9) is *Institutional ownership*, which is the fraction of the firm's shares owned by financial institutions, which we obtain from 13F filings. The additional independent variable in column (10) is *Tobin's Q*, defined as assets minus cash and cash equivalent securities plus book value on equity scaled by assets. We scale up the dependent variable, SCRisk, by a factor of 1000 for readability. The unit of observation is a firm-year. Industries are classified at the two-digit SIC code level. All variables are defined in Internet Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table 3
Supply Chain Risk and the Composition of Supply Chains.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Number of suppliers	Number of suppliers in the same continent	Number of U.S. suppliers	Number of suppliers in different continents	Number of industry leader suppliers	Number of suppliers	Number of suppliers in the same continent	Number of U.S. suppliers	Number of suppliers in different continents	Number of industry leader suppliers
	OLS	OLS	OLS	OLS	OLS	IV	IV	IV	IV	IV
SCRisk	0.0190*** (0.0068)	0.0119*** (0.0041)	0.0098*** (0.0035)	0.0035 (0.0022)	0.0053* (0.0029)	2.8462*** (1.0807)	1.1130** (0.4700)	0.9085** (0.3913)	0.6007** (0.2431)	1.1666** (0.4539)
SCSentiment	0.0010 (0.0008)	0.0006 (0.0004)	0.0006 (0.0004)	0.0001 (0.0002)	0.0005 (0.0003)	−0.0047 (0.0041)	−0.0016 (0.0017)	−0.0012 (0.0014)	−0.0011 (0.0009)	−0.0018 (0.0017)
Size	2.3134*** (0.2403)	1.2159*** (0.1191)	1.1632*** (0.1114)	0.4395*** (0.0592)	1.1577*** (0.1062)	2.2296*** (0.4846)	1.1832*** (0.1989)	1.1365*** (0.1689)	0.4218*** (0.1092)	1.1232*** (0.1969)
Tobin's Q	0.0619 (0.0676)	−0.0239 (0.0335)	0.0155 (0.0303)	0.0070 (0.0181)	0.0352 (0.0287)	0.3977* (0.2072)	0.1069 (0.0871)	0.1223* (0.0724)	0.0780* (0.0451)	0.1732** (0.0855)
Cash holdings	0.2383 (0.6254)	−0.6564* (0.3545)	−0.6254* (0.3342)	0.0694 (0.1743)	−0.5957** (0.2964)	−4.3493* (2.6376)	−2.4431** (1.1043)	−2.0838** (0.9115)	−0.8998 (0.5794)	−2.4801** (1.0968)
Cash flow	−1.9754*** (0.3655)	−1.1778*** (0.2190)	−1.1711*** (0.2033)	−0.4918*** (0.1122)	−1.0049*** (0.1714)	−1.3252 (1.2305)	−0.9247* (0.4991)	−0.9644** (0.4182)	−0.3544 (0.2754)	−0.7378 (0.4983)
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Ind x year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	24,139	24,139	24,139	24,139	24,139	24,139	24,139	24,139	24,139	24,139
Adj. R-squared	0.8268	0.7533	0.7625	0.6594	0.8317	–	–	–	–	–
First-stage coeff	–	–	–	–	–	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)
First-stage F-stat	–	–	–	–	–	12.916	12.916	12.916	12.916	12.916

Notes: This table reports estimates of the effects of a firm's SCRisk and SCSentiment at $t-1$ on its number of suppliers at t . The dependent variables in columns (1) and (6), (2) and (7), (3) and (8), (4) and (9), and (5) and (10) are the number of suppliers, the number of suppliers in the same continent as the firm, the number of U.S. suppliers, the number of suppliers in different continents, and the number of suppliers that are industry leaders, respectively. Industry leaders are defined as suppliers with sales above the median of their 3-digit SIC industry. Firm controls include size, Tobin's Q, cash holdings, and cash flow at $t-1$. The unit of observation is a firm-year. Industries are classified at the two-digit SIC code level. Columns 1–5 report OLS regression results. Columns 6–10 use the supply chain risk of its suppliers as an instrument. All variables are defined in Internet Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

and multisourcing key inputs to reduce their dependence on specific suppliers. In column 1, firms indeed appear to increase the number of suppliers in response to supply chain risk: going from the median to the 95th percentile of SCRisk leads to a 3.6% increase in the number of suppliers.

Firms address supply chain risk not only through the quantity but also through the quality of their suppliers. Elliott et al. (2022) argue that firms can minimize the probability of production disruption through the reliability of the supplier network or by forming stronger relationships with their suppliers. One way to do this is through geographical proximity. First, uncertainty about transportation costs or travel damages increases as the physical distance between a firm and its supplier increases (Schmitt and Van Biesebroeck, 2013; Bray et al., 2019). Second, firms can better monitor physically closer plants, which can help them maintain consistent product quality (Giroud, 2013). Finally, better monitoring coupled with more face-to-face communication can help firms build trust with their suppliers (Schmitt and Van Biesebroeck, 2013). Therefore, we would expect that firms establish relationships with geographically closer suppliers when supply chain risk increases.

We test this conjecture in columns 2 and 3. We look at the number of suppliers in the same continent and in the U.S., respectively. The estimated coefficients on SCRisk are positive and statistically significant at the 1% confidence level and imply that following an increase in supply chain risk, U.S. public firms start working with an increasing number of suppliers located in the same continent, mainly in the U.S. In column 4, we see no change in the number of suppliers in different continents, indicating that nearshoring increases the diversification of a firm's supply chains but does not necessarily lead to more fragmentation of the supply chains.

Another way to increase the reliability of the supplier network is to work with suppliers that are leaders in their industries. Industry leaders are expected to have a reputation of delivering on their commitments, which should be particularly important when firms have concerns about the ability to source their inputs. We test this conjecture in column 5. We define firms as industry leaders if their sales are above the median within their 3-digit industry. The positive coefficient of interest shows that firms establish relationships with suppliers that are leaders in their industries when supply chain risk increases.

Taken together, these findings show that firms strategically choose their suppliers to minimize the risk of costly production disruption (Elliott et al., 2022). Supply chain diversification and nearshoring (reliance on close suppliers) thus appear complementary mechanisms to address supply chain risk.

Interestingly, decreases in SCSentiment are not associated with greater supply chain diversification, suggesting that if anything, shocks to the conditional mean have different effects from supply chain uncertainty. Firms appear to expand the number of close suppliers when they experience positive supply chain shocks, possibly to increase production when facing favorable conditions.

We also address the concern that discussions of supply chain risk could be endogenous. In particular, companies that are experiencing difficulties and have to switch suppliers, possibly resulting in higher costs, could strategically discuss supply chain uncertainty even if their problems do not derive from the supply chain. We conjecture that strategic discussions of supply chain risk are less likely if a firm's suppliers are also discussing supply chain risk. We thus use as an instrument for a firm's SCRisk the maximum SCRisk of all its suppliers (whether domestic or international) that we observe from Factset Revere and for which we can obtain earnings conference calls from Eikon and Capital IQ. By considering the maximum SCRisk of a firm's suppliers, we take into account that inputs are complements, and a firm's production is likely to be disrupted even if only one of the suppliers experiences problems.

Columns 6 to 10 report the instrumental variable estimates. The effect of the instrument on a firm's SCRisk in the first stage is positive and significant, as we expect.⁵ Furthermore, the F-statistic of the excluded instrument indicates that our instrument is not weak. The second stage estimates continue to support our earlier findings that following increases in supply chain risk, firms increase their number of suppliers, relying especially on suppliers that are in the same continent or domestic and are industry leaders. Importantly, it appears that firms also add suppliers in other continents, even though to a lower extent, confirming that supply chain uncertainty does not lead to more supply chain fragmentation.

Importantly, the coefficient on SCRisk is much larger in the IV estimates. For instance, in column 6, going from the median to the 90th percentile of SCRisk leads to a 67% increase in the number of suppliers. The larger parameter estimate in the instrumental variable regressions is likely to depend on two factors. First, SCRisk contains measurement error. To the extent that the measurement error is uncorrelated with firms' supply chain characteristics and just depends on idiosyncrasies of the conference calls languages, it may drive down the coefficient estimates in the ordinary least squares regressions. Second, and more importantly, when we use the SCRisk of the suppliers as an instrument, we are likely to capture disproportionately large increases in SCRisk, which warrant discussions by several firms in the supply chain. Thus, the local average treatment effect that we capture in the instrumental variable estimate is necessarily larger.

5.2. Vertical integration

Supply chain risk can affect not only the composition of suppliers but also the firms' boundaries. Theories of the boundaries of the firm suggest that supply assurance concerns may lead firms to integrate vertically (Williamson, 1971; Bolton and Whinston, 1993). When supply chain uncertainty increases and bottlenecks become more likely, the intensity of hold up problems between a firm and its

⁵ Table IA.10 reports the full first-stage estimates. In Table IA.11, our results are robust if we instrument for both SCRisk and SCSentiment using the SCRisk and SCSentiment of the firms' suppliers. However, the instruments are weak as the F-statistic of the excluded instruments is always around 2. For this reason, we present these estimates in the Internet Appendix. The consistency of the parameter estimates between ordinary least squares and alternative instrumentation strategies assuages concerns that biases may drive our estimates. Table IA.12 shows that both ordinary least squares and instrumental variable estimates are robust if we do not control for SCSentiment.

suppliers intensifies, making vertical integration more desirable (Grossman and Hart, 1986). For these reasons, we expect firms experiencing heightened supply chain risk to have a higher probability of acquiring firms in an upstream industry to better control the access to inputs. We expect the probability of the acquisition of firms in downstream industries to increase as well because the identity of the acquirer and the target depends on technological reasons affecting their relative size, their liquid holdings, and access to external finance. For a firm to integrate with a supplier, it may just be optimal to become a target because, theoretically, what matters is the common ownership of different stages of the production process.

Table 4 explores whether firms engage in more vertical M&As when supply chain risk increases. Throughout the analysis, we include interactions of two-digit SIC industry and year fixed effects to control for industry shocks, which are known to lead to merger waves (e.g., Ahern and Harford, 2014). We also include firm fixed effects thus capturing how changes in supply chain risk change a firm's propensity to vertically integrate. We report ordinary least squares (columns 1 to 3) and instrumental variable (columns 4 to 6) specifications, in which we use the maximum SCRisk of a firm's suppliers as the instrument.

Column 1 evaluates the probability that a firm is involved in an M&A with a firm in an upstream industry, while column 2 considers M&As with firms in downstream industries. In both cases, we find that an increase in SCRisk leads to a higher probability of M&As. In particular, a one-standard-deviation increase in SCRisk increases the probability of an M&A with a supplier or a customer by 5.9% and 8.1% relative to the baseline merger probability of 10.0% and 7.2%, respectively. In column 3, we find that firms engage in M&As in industries that are not related to a lower extent when supply chain risk increases, indicating that we are not just capturing firms' general propensity to make acquisitions.

In columns 4 to 6, we address the concern that firms discuss supply chain risk because they wish to integrate vertically even if they are not experiencing supply chain uncertainty. The estimated effects of SCRisk on firms' propensity to vertically integrate are significantly larger in the instrumental variable estimates, suggesting that episodes of large increases in supply chain risk, as those that we capture when we exploit variation due to the SCRisk of the suppliers, are followed by a much higher propensity of firms to vertically integrate. In particular, a one-standard-deviation increase in SCRisk increases the probability of an M&A with a supplier or a customer by 25.6% and 31.6%, respectively. In addition, Table IA.13 shows that both ordinary least squares and instrumental variable estimates are robust if we do not control for SCSentiment.

So far, we have defined upstream and downstream industries considering input-output tables. Specifically, any industry that exhibits a bilateral flow is considered vertically related. In this way, we capture that limited availability of any input can cause bottlenecks in the presence of production complementarities. One may think however that inputs that are larger proportions of an industry's costs of production matter most. In Table IA.14, we repeat our tests considering as vertically related only upstream (downstream) industries that account for at least 1% of purchases (sales) as Garfinkel and Hankins (2011) do to define vertical M&As. Our results are qualitatively invariant.

Table 4
Supply Chain Risk and Vertical M&As.

	(1)	(2)	(3)	(4)	(5)	(6)
	M&A with supplier	M&A with customer	Unrelated M&As	M&A with supplier	M&A with customer	Unrelated M&As
	OLS	OLS	OLS	IV	IV	IV
SCRisk	0.0007** (0.0003)	0.0007** (0.0003)	−0.0002* (0.0001)	0.0614** (0.0268)	0.0550** (0.0242)	−0.0174* (0.0105)
SCSentiment	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	−0.0001 (0.0001)	−0.0001 (0.0001)	0.0000 (0.0000)
Size	−0.0205*** (0.0062)	−0.0160*** (0.0050)	−0.0011 (0.0044)	−0.0223** (0.0111)	−0.0176* (0.0095)	−0.0006 (0.0049)
Tobin's Q	0.0039* (0.0022)	0.0052*** (0.0018)	0.0008 (0.0015)	0.0111** (0.0049)	0.0117*** (0.0044)	−0.0012 (0.0021)
Cash holdings	0.1001*** (0.0237)	0.0708*** (0.0199)	0.0512*** (0.0187)	0.0016 (0.0649)	−0.0174 (0.0575)	0.0790*** (0.0288)
Cash flow	0.1128*** (0.0141)	0.0769*** (0.0116)	0.0460*** (0.0094)	0.1268*** (0.0297)	0.0894*** (0.0263)	0.0421*** (0.0120)
Firm FE	Y	Y	Y	Y	Y	Y
Industry x year FE	Y	Y	Y	Y	Y	Y
Observations	24,139	24,139	24,139	24,139	24,139	24,139
Adjusted R-squared	0.1379	0.1177	0.1172	—	—	—
First-stage coeff	—	—	—	0.0100*** (0.0037)	0.0100*** (0.0037)	0.0100*** (0.0037)
First-stage F-stat	—	—	—	12.916	12.916	12.916

Notes: This table reports estimates of the effects of SCRisk at $t-1$ on the probability that a firm is involved in M&As at t . The dependent variables in columns (1) and (4) and (2) and (5) are *M&A with supplier* and *M&A with customer*, which are indicator variables that equal one if the firm conducts an M&A with a firm in an upstream or downstream industry, respectively. The dependent variable in columns (3) and (6) is *Unrelated M&A*, which is an indicator variable that equals one if the firm conducts an M&A with a firm that is not in an upstream, downstream, or the same industry. The unit of observation in each regression is a firm-year. Firm controls include size, Tobin's Q, cash holdings, and cash flow at $t-1$. Industries are classified at the two-digit SIC code level. Columns 1–3 report OLS regression results. Columns 4–6 use the supply chain risk of its suppliers as an instrument. All variables are defined in Internet Appendix A. Robust standard errors clustered by firm are in parentheses. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

Table 5 tests another implication of the vertical integration theories. Vertical integration is expected to generate a larger surplus when supply assurance concerns are severe. Therefore, we expect the announcement of a vertical M&A to generate higher abnormal returns when firms are experiencing high supply chain risk. This is precisely what we find when we regress a firm's cumulative abnormal returns (CAR), obtained by estimating the market model over a $[-255, -31]$ day estimation window and cumulating abnormal returns over a $[-1, +1]$ window around merger announcement, on the interaction of SCRisk and an indicator for vertical mergers. In column 1, for example, the coefficient on the interaction term is positive and significant. The effect is also economically large. When SCRisk increases from the 50th percentile to the 90th percentile, the vertical merger CAR increases by 0.23 percentage point, representing a 23% increase in CAR compared to the mean of 1%. In columns 2 and 3, we saturate the model with firm level controls and industry and year fixed effects that could have an independent effect on the value created by a merger. Our results are qualitatively invariant. To further corroborate our interpretation of the empirical evidence that supply chain risk leads to more vertical integration, we also exploit that financial constraints prevent companies from engaging in vertical M&As. If SCRisk captures an actual increase in supply chain uncertainty, we should observe that its effect on the probability to vertically integrate is reduced for financially constrained firms. Put differently, financially constrained firms should be affected as much as other firms by supply chain risk but should be less able to react to it.

In Table IA.15, we use two measures of financial constraints to test this conjecture: The index based on size and age introduced by [Hadlock and Pierce \(2010\)](#) in Panel A and the Whited-Wu index proposed by [Whited and Wu \(2006\)](#) in Panel B. We classify firms as financially constrained (unconstrained) if their index value is above (below) the median. In columns 1 and 2 of Panels A and B, the estimated coefficients on the interaction terms are negative and statistically significant, which shows that financially constrained firms are significantly less likely to be involved in vertical M&As when supply chain risk increases.

5.3. Robustness

Table IA.16 shows that the results in [Table 3](#) and [Table 4](#) are robust to the inclusion of the political and climate risk proxies developed by [Hassan et al. \(2019\)](#) and [Sautner et al. \(2023\)](#), respectively. Furthermore, contrary to supply chain risk, political and climate risks have a negative or no effect on vertical integration and supplier composition. Not only does this confirm that SCRisk captures a different source of risk, but also that it has considerably different effects on corporate policies and industrial structure. In Table IA.17, we address the concern that supply chain risk could capture general negative conditions. However, our results are invariant when we control for the overall sentiment of a firm's earnings call.

Another potential concern is that SCRisk is measured with noise from the earnings call transcripts. While using the SCRisk of a firm's suppliers helps to address this concern, we also exploit that our topics analysis identifies discussions of supply chain risks in the context of general financial and analyst-related and liquidity issues that are typical of earnings conference calls. These snippets might reflect noise. We consider supply chain risk to be measured with noise for a firm if during a year >75% of the snippets that incorporate supply chain risk discussions are associated with a probability of discussing general financial and analyst and liquidity issues in the top quartile. We either substitute SCRisk that we identify as potential noise with the two-digit SIC industry median SCRisk during the same

Table 5
Stock Market Reaction to Vertical M&A Announcements.

	(1)	(2)	(3)
	CAR $[-1, +1]$	CAR $[-1, +1]$	CAR $[-1, +1]$
SCRisk	-0.0004*** (0.0001)	-0.0004** (0.0001)	-0.0004*** (0.0002)
Vertical merger	-0.0004 (0.0024)	-0.0022 (0.0024)	-0.0028 (0.0030)
SCRisk * Vertical merger	0.0004** (0.0002)	0.0004** (0.0002)	0.0004** (0.0002)
Size		-0.0041*** (0.0006)	-0.0045*** (0.0006)
Tobin's Q		-0.0017** (0.0008)	-0.0012 (0.0009)
Cash holdings		-0.0287*** (0.0074)	-0.0231*** (0.0070)
Cash flow		-0.0158 (0.0119)	-0.0161 (0.0126)
Year FE	N	N	Y
Industry FE	N	N	Y
Observations	7668	7668	7668
Adjusted R-squared	-0.0001	0.0085	0.0138

Notes: This table reports OLS regression results for firms' cumulative abnormal returns (CAR) around M&A announcements. The dependent variable is the CAR over a three-day event window $[-1, +1]$ around an M&A announcement, obtained by estimating the market model over a $[-255, -31]$ day estimation window. Vertical merger is a dummy variable that equals one if the firm conducts an M&A with a firm from an upstream or a downstream industry and zero if the firm conducts an M&A with a firm from neither an upstream nor a downstream industry. The unit of observation in each regression is at the deal level. Firm controls include size, Tobin's Q, cash holdings, and cash flow at $t-1$. All variables are defined in Internet Appendix A. Statistical significance at the 1%, 5%, and 10% level is denoted by ***, **, and *, respectively.

year or drop the observations. Our results in Table IA.18 and IA.19 are unaffected.

6. Conclusions

Supply chains and input availability are crucial determinants of comparative advantage. We develop a novel proxy for supply chain uncertainty based on textual analysis and explore how supply chain uncertainty affects corporate policies. We show that firms facing more supply chain uncertainty diversify their suppliers by establishing new relationships. Firms also establish relationships with suppliers in the same continent and suppliers that are industry leaders but do not decrease their reliance on suppliers in other continents. More importantly, firms that face more supply chain risk are more likely to become vertically integrated by entering into M&As with their customers and suppliers.

Our results suggest that higher supply chain uncertainty could be associated with changes in the geography and organization of economic activity with protracted long-term effects on the performance of different geographical areas. Exploring these issues is an exciting area for future research.

Declaration of Competing Interest

Nuri Ersahin declares that he has no relevant conflicts or material financial interests that relate to the research described in this paper.

Mariassunta Giannetti declares that she has no relevant conflicts or material financial interests that relate to the research described in this paper.

Ruidi Huang declares that he has no relevant conflicts or material financial interests that relate to the research described in this paper.

Data availability

[ReplicationPackage_EGH.zip \(Reference data\)](#) (Mendeley Data)

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Appendix A. Supplementary data

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References

- Acemoglu, D., Carvalho, V.M., Ozdaglar, A., Tahbaz-Salehi, A., 2012. The network origins of aggregate fluctuations. *Econometrica* 80, 1977–2016.
- Ahern, K., Harford, J., 2014. The importance of industry links in merger waves. *J. Financ.* 69, 527–576.
- Alfaro, L., Antràs, P., Chor, D., Conconi, P., 2019. Internalizing global value chains: a firm-level analysis. *J. Polit. Econ.* 127, 508–559.
- Alfaro, L., García-Santana, M., Moral-Benito, E., 2021. On the direct and indirect real effects of credit supply shocks. *J. Financ. Econ.* 139, 895–921.
- Antràs, P., Chor, D., 2013. Organizing the global value chain. *Econometrica* 81, 2127–2204.
- Antràs, P., Chor, D., 2022. Global value chains. *Handb. Int. Econ.* 5, 297–376.
- Aral, K.D., Giambona, E., Lopez Aliouchkin, R., Phillips, G.M., 2021. Global Supply Chain Disruptions. Working Paper.
- Baker, S., Bloom, N., Davis, S.J., 2016. Measuring economic policy uncertainty. *Q. J. Econ.* 131, 1593–1636.
- Baker, S., Bloom, N., Terry, S.J., 2023. Using disasters to estimate the impact of uncertainty. *Rev. Econ. Stud.* (forthcoming).
- Barrot, J.-N., Sauvagnat, J., 2016. Input specificity and the propagation of idiosyncratic shocks in production networks. *Q. J. Econ.* 131, 1543–1592.
- Bena, Jan, Li, K., 2014. Corporate innovations and mergers and acquisitions. *J. Financ.* 69, 1923–1960.
- Benigno, G., di Giovanni, J., Groen, J.J.J., Noble, A.I., 2022. A new barometer of global supply chain pressures. *Lib. Street Econ.* January 4 <https://libertystreeteconomics.newyorkfed.org/2022/01/a-new-barometer-of-global-supply-chain-pressures/>.
- Berger, D., Dew-Becker, I., Giglio, S., 2020. Uncertainty shocks as second-moment news shocks. *Rev. Econ. Stud.* 87, 40–76.
- Bimpikis, K., Fearing, D., Tahbaz-Salehi, A., 2018. Multisourcing and miscoordination in supply chain networks. *Oper. Res.* 66, 1023–1039.
- Bloom, N., 2014. Fluctuations in uncertainty. *J. Econ. Perspect.* 28, 153–176.
- Bolton, P., Whinston, M.D., 1993. Incomplete contracts, vertical integration, and supply assurance. *Rev. Econ. Stud.* 60, 121–148.
- Bonadio, B., Huo, Z., Levchenko, A., Pandalai-Nayar, N., 2021. Global supply chains in the pandemic. *J. Int. Econ.* 133, 1–23.

- Bozanic, Z., Roulstone, D.T., Buskirk, A.V., 2018. Management earnings forecasts and other forward-looking statements. *J. Account. Econ.* 65, 1–20.
- Bray, R., Colak, A., Serpa, J., 2019. Supply chain proximity and product quality. *Manag. Sci.* 65, 4079–4099.
- Caldara, D., Iacoviello, M., 2022. Measuring geopolitical risk. *Am. Econ. Rev.* 112, 1194–1225.
- Carvalho, V.M., Nirei, M., Saito, Y.U., Tahbaz-Salehi, A., 2021. Supply chain disruptions: evidence from the great East Japan earthquake. *Q. J. Econ.* 136, 1255–1321.
- Choi, T.Y., Rogers, D., Vakil, B., 2020. Coronavirus is a wake-up call for supply chain management. *Harv. Bus. Rev.* 27, 364–398.
- Chopra, S., Meindl, P., 2016. *Supply Chain Management: Strategy, Planning, and Operation*, Sixth ed. Prentice Hall, Upper Saddle River, N.J.
- Coase, R., 1937. The nature of the firm. *Econometrica* 4, 386–405.
- Costello, A.M., 2020. Credit market disruptions and liquidity spillover effects in the supply chain. *J. Polit. Econ.* 128, 3434–3468.
- Crosignani, M., Macchiavelli, M., Silva, A., 2023. Pirates without borders: the propagation of cyberattacks through firms' supply chains. *J. Financ. Econ.* 147, 432–448.
- Dasgupta, S., Harford, J., Ma, F., Wang, D., Xie, H., 2020. Mergers under the Microscope: Analysing Conference Call Transcripts. Working Paper.
- Elliott, M., Golub, B., Leduc, M.V., 2022. Supply network formation and fragility. *Am. Econ. Rev.* 112, 2701–2747.
- Ersahin, N., Giannetti, M., Huang, R., 2023. Trade Credit and the Stability of Supply Chains. Working Paper.
- Fan, J., Goyal, V., 2006. On the patterns and wealth effects of vertical mergers. *J. Bus.* 79, 877–902.
- Florackis, C., Louca, C., Michael, R., Weber, M., 2023. Cybersecurity risk. *Rev. Financ. Stud.* 36, 351–407.
- Frésard, L., Hoberg, G., Phillips, G.M., 2020. Innovation activities and integration through vertical acquisitions. *Rev. Financ. Stud.* 33, 2937–2976.
- Garfinkel, J.A., Hankins, K.W., 2011. The role of risk management in mergers and merger waves. *J. Financ. Econ.* 101, 515–532.
- Giroud, X., 2013. Proximity and investment: evidence from plant-level data. *Q. J. Econ.* 128, 861–915.
- Grossman, S., Hart, O., 1986. The costs and benefits of ownership: a theory of vertical and lateral integration. *J. Polit. Econ.* 94, 691–719.
- Hadlock, C.J., Pierce, J.R., 2010. New evidence on measuring financial constraints: moving beyond the KZ index. *Rev. Financ. Stud.* 23, 1909–1940.
- Hassan, T.A., Hollander, S., van Lent, L., Tahoun, A., 2019. Firm-level political risk: measurement and effects. *Q. J. Econ.* 134, 2135–2202.
- Hassan, T.A., Hollander, S., van Lent, L., Tahoun, A., 2020. Firm-level exposure to epidemic diseases: Covid-19, SARS, and H1N1. *Review of Financial Studies* 36, 4919–4964.
- Hassan, T.A., Hollander, S., van Lent, L., Tahoun, A., 2020. The global impact of Brexit uncertainty. *Journal of Finance*, forthcoming.
- Hertzel, M.G., Li, Z., Officer, M.S., Rodgers, K.J., 2008. Inter-firm linkages and the wealth effects of financial distress along the supply chain. *J. Financ. Econ.* 87, 374–387.
- Hsu, P., Li, K., Liu, X., Wu, H., 2022. Consolidating product lines via mergers and acquisitions: evidence from the USPTO trademark data. *J. Financ. Quant. Anal.* 1–25.
- Klein, B., Crawford, R.G., Alchian, A.A., 1978. Vertical integration, appropriable rents, and the competitive contracting process. *J. Law Econ.* 21, 297–326.
- Kopytov, A., Mishra, B., Nimark, K., Taschereau-Dumouchel, M., 2022. Endogenous Production Networks under Uncertainty. Working Paper.
- Kremer, M., 1993. The O-ring theory of economic development. *Q. J. Econ.* 108, 551–575.
- Li, K., Liu, X., Mai, F., Zhang, T., 2021a. The role of corporate culture in bad times: evidence from the COVID-19 pandemic. *J. Financ. Quant. Anal.* 56, 2545–2583.
- Li, K., Mai, F., Shen, R., Yan, X., 2021b. Measuring corporate culture using machine learning. *Rev. Financ. Stud.* 34, 3265–3315.
- Li, Q., Shan, H., Thang, Y., Yao, V., 2023. Corporate climate risk: measurements and responses. *Rev. Financ. Stud.* forthcoming.
- Libby, R., Libby, P., Hodge, F., 2020. *Financial Accounting*. McGraw-Hill/Irwin.
- Loughran, T., McDonald, B., 2011. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *J. Financ.* 66, 35–65.
- Manning, C., Raghavan, P., Schütze, H., 2008. *Introduction to Information Retrieval*. Cambridge University Press, Cambridge.
- Sautner, Z., van Lent, L., Vilkov, G., Zhang, R., 2023. Firm-level climate change exposure. *J. Financ.* 78, 1449–1498.
- Schmitt, A., Van Biesebroeck, J., 2013. Proximity strategies in outsourcing relations: the role of geographical, cultural and relational proximity in the European automotive industry. *J. Int. Bus. Stud.* 44, 475–503.
- Song, M., Wu, Y.B., 2008. Handbook of research on text and web mining technologies.
- Whited, T.M., Wu, G., 2006. Financial constraints risk. *Rev. Financ. Stud.* 19, 531–559.
- Williamson, O.E., 1971. The vertical integration of production: market failure considerations. *Am. Econ. Rev.* 61, 112–123.